

How does this argument become closer to the base case? The range halves in size for each recursive call, so it eventually becomes empty.

The following `quicksort()` function in the *quicksort.py* Python program sorts the values in the `items` list into ascending order:

```
def quicksort(items, left=None, right=None):
    # By default, `left` and `right` span the entire range of `items`:
    if left is None:
        left = 0 # `left` defaults to the 0 index.
    if right is None:
        right = len(items) - 1 # `right` defaults to the last index.

    print('\nquicksort() called on this range:', items[left:right + 1])
    print('.....The full list is:', items)

    if right <= left: ❶
        # With only zero or one item, `items` is already sorted.
        return # BASE CASE

    # START OF THE PARTITIONING
    i = left # i starts at the left end of the range. ❷
    pivotValue = items[right] # Select the last value for the pivot.

    print('.....The pivot is:', pivotValue)

    # Iterate up to, but not including, the pivot:
    for j in range(left, right):
        # If a value is less than the pivot, swap it so that it's on the
        # left side of `items`:
        if items[j] <= pivotValue:
            # Swap these two values:
            items[i], items[j] = items[j], items[i] ❸
            i += 1

    # Put the pivot on the left side of `items`:
    items[i], items[right] = items[right], items[i]
    # END OF THE PARTITIONING

    print('...After swapping, the range is:', items[left:right + 1])
    print('Recursively calling quicksort on:', items[left:i], 'and', items[i + 1:
right + 1])

    # Call quicksort() on the two partitions:
```